

# HERMANN PAUWELLS RIVERA

## Software Engineer | AI/ML · Full-Stack

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### PROFILE

Software engineer focused on AI/ML: Python (TensorFlow, FastAPI) and full-stack TypeScript (React/Next.js, Node.js); final semester of a B.S. in Computer Science and Technology at Tecnológico de Monterrey, available full-time from January 2027. Founder of three early-stage startups built around LLMs and TypeSafe's Jev: AI-agent coordination, nutrition-label analysis and automated video production. Neural-network models for business cases (Datathon 2025).

### TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, Kotlin, C++  
**AI & ML:** TypeSafe's Jev, LLMs (OpenRouter), TensorFlow, faster-whisper, Q-learning, AgentPy  
**Backend & Data:** FastAPI, Flask, Node.js, PostgreSQL, Supabase, SQLite  
**Frontend & Tools:** React, Next.js, Vite, Git, GitHub Actions, FFmpeg, Unity

### EXPERIENCE

**Full-Stack Developer (freelance), Juguetería El Arbolito** Jun 2026 to present

- Built and shipped to production the online store and admin panel for a single-branch toy store with Next.js, TypeScript and Supabase (catalog, cart, checkout, backend).
- Integrated Mercado Pago payments and an LLM chatbot (OpenRouter); prepared the load of the 2,395-product catalog exported from the point of sale (Elevanta) and the online/in-store inventory sync (in progress).

### STARTUPS (FOUNDER)

**HIVEMIND** Co-founder · team chat for people and AI agents · private pilot Sep 2026 to present

- Android (Kotlin/Jetpack Compose) and Windows (Electron) apps over a self-hosted ntfy server; HMAC-SHA256 signed messages, single-use invitations and 104 automated tests.
- Agent orchestration with TypeSafe's Jev (via OpenRouter): a Python dispatcher decides which AI agent takes each message or task (under US\$0.0001 per message, US\$0.50 daily cap), through our own gateway hosted on an Android phone.
- Jev-guided context compaction for long-running Claude Code bots (MIT fork of fast-jev-compaction with a token budget, decay and a lossless archive): ~57K tokens kept instead of ~229K, in 1.5 s.

**recaps-studio** Co-founder · AI narrated-video factory (novel → manhwa-style video) Sep 2026 to present

- Resumable Python pipeline with hash-based caching in SQLite: script, TTS with word-level timing (faster-whisper), shot planning, AI images with a vision judge, subtitles and FFmpeg/Remotion rendering.
- Human approval gates, a credit budget cap, automatic QA and pytest with mocked providers.

**Chécalo** Founder · nutrition app for Mexico · in development Sep 2026 to present

- Scans barcodes (Open Food Facts) and computes Mexico's NOM-051 warning labels with deterministic rules checked against the official gazette (DOF); JavaScript PWA and Android app (Kotlin, CameraX, ML Kit).
- TypeSafe's Jev gives an explainable verdict and flags contradictions between ingredients and the nutrition table; every number comes from the rules. Per-product cached proxy: AI cost grows with the catalog, not with users.

### SELECTED PROJECTS

**CrediFiel Challenge (Datathon 2025)** Data / ML · team project 2025

- Neural-network models (regression + classification) to optimize Credifiel's direct-debit collections: minimize commission costs or maximize recoveries.
- Real financial dataset of direct-debit collections and efficiency metrics (KPIs); Flask API to serve inferences and a React + Vite frontend.

**Multi-agent warehouse simulation (RL)** Python · AgentPy · Unity · FastAPI 2024

- Multi-agent Q-learning with a blackboard (cell reservations, swap blocking) and Contract-Net task auctions: +105% deliveries per episode vs greedy allocation (20 seeded episodes).
- Animated Unity visualization and a FastAPI server.

Other projects: Lumina Reservations (WorkHub MTY) · folk-park · HeatShield · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

### EDUCATION

**B.S. in Computer Science and Technology (ITC), Tecnológico de Monterrey, Monterrey Campus** 2021 to 2026  
Expected graduation: December 2026.

### AWARDS & ACTIVITIES

- Winner, Sustainability category, at Swift Challenge Fest 2025 (Tecnológico de Monterrey), with the HeatShield iOS app.
- SwiftTec Coding Community (2025): hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish (native) · English (C1)